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Research paper

Hybrid CMNV2: DeepFake faces classification and recognition using deep learning methods

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ABSTRACT

Deepfake detection has become a critical component of digital forensics and security, as manipulated images and videos increasingly threaten trust in visual media. However, existing methods often struggle with robustness under post-processing operations such as JPEG compression, Gaussian blur, scaling, and filtering, and with the growing diversity and realism of face image modification (FIM) forgeries. This work proposes CMNV2, a hybrid architecture that integrates MobileNetV2 with a custom CAFFE block to enhance feature extraction and classification accuracy. By adding five additional layers to a pre-trained structure, the model demonstrates superior resilience against complex real-world conditions and achieves 99.10% accuracy across multiple datasets, outperforming 13 baseline CNN models. The study trained and tested CMNV2 on 5,000 images (real and deepfake faces), using a combination of deep neural networks (DNNs), transfer learning (TL), and deep learning (DL) techniques. Compared to 13 CNN-based architectures, the proposed model achieved superior performance across some important evaluation metrics, including accuracy, precision, recall, F1-score, error rate, and computational efficiency. These results highlight hybrid CMNV2 as a robust and efficient solution for deepfake face detection and classification, with potential applications in security, healthcare, and education.

Introduction

Deepfake classification systems using photos or videos are increasingly widespread [1,2]. Deepfake classification models verify identification using facial traits [3,4]. Technologies including Joint Photographic Experts Group (JPEG) compression, Gaussian blur (GB), scaling (SC), median filtering (MED), and mean filtering (ME) are often used in the post-processing of face photos [2,5]. Pre-trained detectors often fail or lose accuracy when face images undergo unknown post-processing [2,4,5]. Therefore, it is becoming more crucial to find out how to detect a huge number of face image modification (FIM) forgeries in complicated settings. A more advanced and flexible technique for identifying fake face photos has to be created. Facial detection is very important in criminal identification, credit card verification and security systems [6]. Biometric authentication faces a significant issue that demands advanced security solutions. One challenge in this field is identifying true features from altered or recreated samples [7]. Liveness-insensitive facial recognition technology can be fooled by modified copies of real people. Face credibility is a problem for camera-based security systems. Due to print errors and image blurriness, current methods cannot always verify a face's authenticity. Advancements in image-capturing and display technologies enable more realistic

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facial pictures [8].

Generative adversarial networks (GANs) can generate realistic visuals, but misuse can be dangerous. GANs can modify personal information in videos, compromising privacy and well-being [9]. Intelligent systems struggle with erroneous face recognition due to the strength of generative models. Training models to detect fake faces becomes less accurate with increasing realistic depiction [10]. Advances in image processing and machine learning make it simpler than ever to develop, modify and produce images of the highest quality. Criminals can apply these technologies to create fake images to cause harm, prevent detection, or mislead algorithms that detect images [11]. Face image GAN models come in 2 categories. The first form, random GAN, creates realistic-looking facial images of imaginary humans. Second, identity-preserving GAN models build realistic images of a target person's face with their input as real image [12]. Identity-preserving GAN paradigm presents challenges in manufacturing fake news and launching targeted malware attacks [3,4].

Social media has seen an abrupt rise in the number of fake avatars, politicians and celebrities. Most of the time, the public is unaware of these appearances, which negatively affects society as a whole [13]. To create detection algorithms, researchers used co-occurrence matrices, pixel points, neural networks, manual feature extraction, and frequency maps. Current GAN techniques, which were based on state-of-the-art image recognition technology, became dominant as a consequence [14]. Modern researchers have developed a number of effective strategies to address various research and development difficulties in response to the quickly advancing technology for producing false images [3,14]. Finding fake images and preventing any damage they can do are the main goals of studying digital image forensics [3]. Algorithms used to identify fake faces often function at the pixel level. When using a convolutional neural network (CNN) model for binary classification in image analysis, this challenge is crucial [15]. In this method, images are first processed via a high-pass filter to prepare them for further processing. Then, the prepared data is used to train a 5-layer neural network. When it comes to identifying fake photos with homology errors, the CNN model works effectively despite its simplicity [16].

Due to advances in technology, it is now simpler than ever to recreate human face images for incorrect media using recordings. These faces could indicate fake attack victims. Numerous commercial and open-source tools can be used to generate appealing videos on certain subjects [17]. This new approach produces fake faces in images, according to studies. Image alteration using synthetic faces raises ethical, legal, and moral issues. Changing an image becomes easy without generating major visual faults [18]. In the last decade, image forensics and manual feature-based forensic systems have been discussed and suggested. The identification of counterfeit images in photo forensics is challenging due to the potential of computer-generated face representations to deceive humans [19]. Although the production of fake videos is not new, DeepFace employs several machine-learning methods to expand its influence [20].

The following are the main contributions of this research:

i. Using 5 different layers, an architectural framework for an improved MobileNetV2 (MNV2) is designed.
ii. With combining improved MobileNetV2 and CAFFE models, we developed a novel hybrid CMNV2 model.
iii. Utilizing a total of 14 different models, we carried out a binary classification task and evaluated the outcomes.
iv. Tables and confusion matrices were used to show the data and findings in an understandable manner.
v. In this study, 14 various models were analyzed and their performance was assessed using 9 different evaluation metrics.

Scope of the Article:

The scope of the article is focused on deepfake face classification using both image and video data through a deep learning-based hybrid model named hybridCMNV2. The study involves:

i. Improving MobileNetV2 through additional layers.
ii. Integrating a CAFFE-based module for faster feature embedding. iii. Applying the model specifically for binary classification of facial images into real vs deepfake.

Objectives of the Article:

i. Design and develop a hybrid deep learning model (CMNV2) for effective and efficient deepfake detection.
ii. Enhance performance by modifying the MobileNetV2 architecture with additional layers and CAFFE block integration.
iii. Benchmark CMNV2 against 13 well-known models using multiple metrics (accuracy, precision, recall, F1-score, error rate, computational time).
iv. Demonstrate the real-world applicability of the model in areas such as intelligent attendance, healthcare, and education.

CMNV2 has previously been applied to mask detection in photo and real-time video streams using five added layers and CAFFE-based detection pipelines [21], and to multi-class real/fake classification on unbalanced datasets with six added layers and CAFFE–Keras/TensorFlow integration [22,23]. In contrast, our study focuses on binary deepfake detection with explicit robustness against post-processing perturbations (compression, blur, scaling), and introduces a distinct five-layer head over frozen MobileNetV2. In the following sections, this article discusses: According to Section 2, for relevant research. This paper provides a detailed discussion of data distribution approaches for training and testing in Section 3. This research study utilizes an improved MobileNetV2 model architecture, as explained in Section 4. Additionally, it shows how proposed method uses the CMNV2 model for detection and classification. Section 5 demonstrates that hybrid CMNV2 technique outperformed 13 others, as shown by a confusion matrix and other distinctive aspects. The study also included deepfake and real face categorisation and prediction accuracy. Section 6 presents a clear summary of the article's conclusions.

Related work

A new method for detecting deepfake faces in digitally modified images or videos, including faces, was suggested. This technique addresses shortcomings of deepfake faces, such as adding a synthetic face area to the original images, using 3D head posture evaluation utilizing face images [24]. A typical counterfeit feature network of GANs is trained using paired learning to distinguish between real and deepfake images [1]. Several studies have been undertaken to detect facial images produced by GAN algorithms due to their potential risk. According to survey research [4], multiple methodologies are utilized to detect an image that is changed or created with GAN model. Various face datasets used for training, evaluating CNN, DNN and incremental classifiers. In the following study, an all-in-one CNN for face analysis [25], a unique idea and complete architecture are presented for integrating face analysis tasks effectively with the processes of face data.

Recent deepfake images and videos modified or manufactured with artificial intelligence (AI) algorithms are more realistic and well-made. These approaches may be used to make fake comments about popular people or videos about imaginary events to influence large audiences to detect deepfakes in web videos [26]. A study explains DeepFace and other face image modification technologies and their detection and classification [27]. A powerful neural network-based image analysis program called Fabricated Face Detect, which was created to recognize and detect many fake or misleading faces. The goal focuses on detecting fake images from humans and GANs [11].

The model was constructed and confirmed using real and deepfake face datasets utilizing deep learning methods based on neural networks

[10,27]. Deepfake images are classified independently using machine learning and deep learning. Support vector machines (SVM) and K-nearest neighbors (KNN) are used for feature vector classification with hyper-parameter modifications [28]. A new fake video detection system was suggested which integrates transfer learning in autoencoders with convolutional and recurrent neural networks (RNN) [29]. A two-stream approach that uses face images and facial information was proposed for a multi-modality problem. A directed attention technique highlights changing regions [10].

The hybrid long short-term memory with extreme machine learning techniques (HLSTM-ELM) system classified the deepfake videos created by a GAN. GAN technique replaces fake data with real images or videos [30]. Deepfake technology can be unsafe, particularly with media coverage. Misinformation and internet abuse are examples of technology misuse that potentially harm society [31]. To address StyleGAN2 and stable diffusion issues for detecting deepfakes in multiclass environments, to evaluate vision transformers (ViTs), a model is created. The challenge is in navigating the dynamic landscape of deepfake development and manipulation [5].

Four key kinds of approaches are identified by DeepFace caused in academia: multimodality-based deepfake detection task (DFDT), image-based, audio-based, and video-based [32]. The research assessed the performance of VGG-19 and ResNet-50 models, pre-trained on face datasets of ImageNet and VGG, for distinguishing real and deepfake face images [33]. To address the problems caused by the limited generalizability of textures and artifacts collected jointly, a texture and artifact detector (TAD) technique was proposed. The research looks at the similarities between fake data, specifically how consistent textures and artifacts might be used to identify synthetic material [34]. Additionally, to aggregate artifact patterns and textural differences among various kinds of counterfeit, a model ensemble learning technique is used, suggesting a threshold of interpretability for identifying deepfake audio that is comparable to human perception [35]. The model applies the explainable artificial intelligence (XAI) image classification algorithm to noise issues, and the method suggests using a data format to enhance interpretability by looking at the human cognitive approach to XAI for image categorization. DL-based techniques, such as ResNet-Swish-Dense54, are used to accurately detect deepfakes. The first step in preprocessing is to remove people's faces from the raw video frame. The ResNet-Swish-Dense54 technique is then used to analyze the extracted face in order to identify if the data is modified or genuine [36].

A newly created approach to data preservation based on BC-based federated learning (FL) is presented by combining a convolutional-deconvolutional capsule network (SEGCAPS) and CNN techniques for visual feature extraction with CapsNet training to enhance generalizability [37]. The DL is more accurate when pretreatment and TL techniques are used, which describes an unconventional method for identifying deepfake videos using face forensic and DFDC databases [38]. Furthermore, video underwent frequency-based frame extraction during the pre-processing phase using CNN-long short-term memory (LSTM) and the fake videos were detected. A deep hybrid system for identifying deepfake films is developed to extract information about face attributes from the videos by using facial landmark detection. A multilayer perception (MLP) is trained using this data to distinguish between genuine and deepfake videos [39].

A study was conducted on the existing image, video and audio deepfake databases to support the development of next-generation detectors with better accuracy, generalization, robustness and explainability. This explores key challenges and future research directions in audio and video deepfake generation and mitigation [2].

Experiment setup

"Ciplab@Yonsei University" developed the original images dataset, which is accessible in the Kaggle repository [40]. This study divides 2,040 human face images into 2 categories. This study covers training and testing of 2 binary classification models 3.1. In this research, 2,040 images are augmented to 5,000 images of 2 classes by flipping 180 degrees horizontally and cropping to increase model training chances.

The dataset used in this study comprises 5,000 facial images, equally divided into 2,500 real and 2,500 deepfake samples. Real images were sourced from publicly available facial datasets such as CelebA and LFW, while deepfake images were obtained from DeepFakeDetection (DFD) and FaceForensics++, ensuring a wide range of manipulation techniques and content diversity. All images were carefully pre-processed with resizing to (224 × 224), grayscale-to-RGB conversion where needed, and normalization. Additionally, we applied data augmentation techniques including rotation, horizontal flipping, brightness adjustment, and zoom to increase diversity and prevent overfitting. The dataset was also manually reviewed to validate labels and maintain class balance, ensuring the fairness and robustness of the binary classification process.

The image selection approach: initial random sampling followed by automated curation. Quality control measures, including:

Minimum image resolution requirement.

Face alignment using facial landmark detection.

Verification of full face visibility and absence of occlusion.

Manual inspection of a subset to confirm automated filtering.

Removal of duplicate or corrupted images.

Images from each dataset were initially randomly sampled to avoid selection bias. An automated curation process using a facial landmark detection algorithm ensured proper alignment. Images not meeting a minimum resolution of (224 × 224) pixels, those with partially occluded faces, or exhibiting significant compression artifacts were excluded. A subset of images was manually inspected to verify quality. Duplicates and corrupted files were removed, resulting in a clean dataset of 5,000 images for training and testing the CMNV2 model.

Training and testing distribution of data

The training data, which helps to learn and improve models' prediction capabilities, is shown in Fig. 1 (a). A total of 80% i.e., 4,000 images, were used to train 2 groups, proving the model's extensive dataset access and complicated pattern detection. There are 2,000 photo images in the 'real face' and 2,000 images of 'deepfake face' photo images, respectively. As can be shown in Fig. 1 (b), 20% i.e., the 1,000 images for testing are used by proposed model in the testing set. This phase is crucial to model accuracy and predictability. A total of 1,000 images were divided into 2 categories: 500 images for 'real face' and 500 images for 'deepfake face'.

Proposed methodology

Building on prior CMNV2 variants for face mask detection and face classification [21–23], we introduce a modified CMNV2 with five additional layers tailored for binary deepfake detection in photo and real-time video settings. Unlike earlier CMNV2 studies that targeted masks or multi-class real/fake categorization, our focus is forensic robustness to post-processing (JPEG compression, Gaussian blur, scaling, filtering) and comparative evaluation against 13 CNN baselines. It should be noted that a DNN structure developed using Keras, TensorFlow, and OpenCV modules accurately classifies and detects real and deepfake faces in real-time video and photos by our technique to classify and detect images. The improvised MobileNetV2 framework with the different layers added is shown in Fig. 2.

The CAFFE module in the proposed CMNV2 model refers to a custom-designed convolutional block inspired by efficient deep learning principles aimed at Fast Feature Embedding (FFE). This module consists of a sequence of depthwise separable convolutions, batch normalization, and ReLU activations, which accelerate computation while preserving discriminative power. The goal is to quickly reduce spatial dimensions and extract relevant patterns that are crucial for distinguishing between

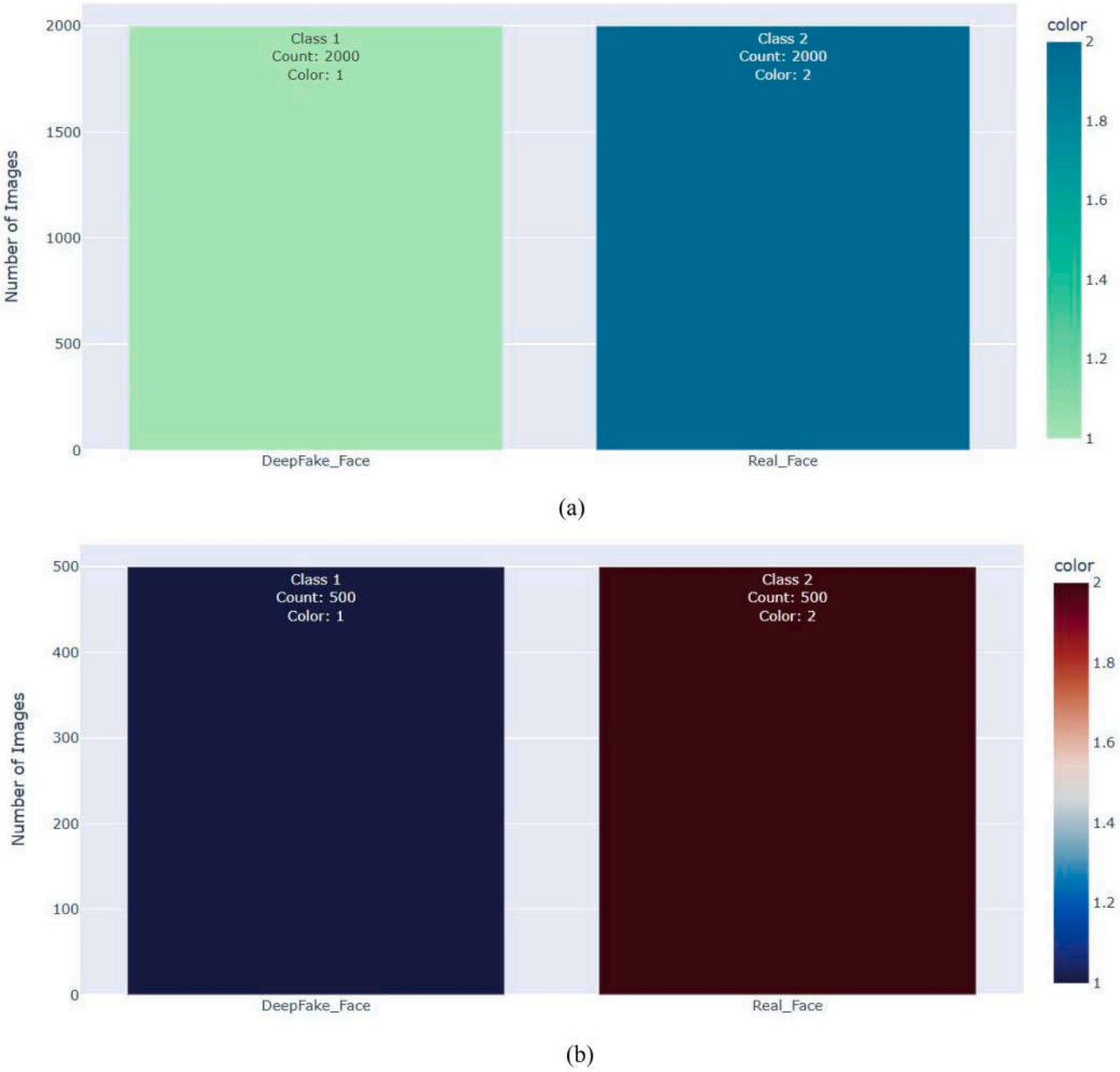


Fig. 1. Visualization of Training and Testing data distribution.

real and deepfake facial features [41]. The CAFFE module was designed as a lightweight extension that optimizes feature extraction by reducing redundant computations in convolutional operations. Specifically, the module employs depthwise separable convolutions combined with group normalization to lower computational cost while maintaining representational capacity. Furthermore, it integrates a fast activation function to minimize latency during inference. To strengthen this discussion, we have incorporated references to prior work on compact convolutional architectures and real-time optimization techniques [21, 22]. The CAFFE module, therefore, not only accelerates training and inference but also enhances compatibility with resource-constrained environments, making it particularly suitable for real-time applications such as surveillance, healthcare, and education.

Improved MobileNetV2

This research study added 5 layers to the MNV2 design while maintaining the original MobileNetV2 layers. Baseline CNN MobileNetV2 architecture [42] was used to build it. We acquired features using transfer learning. Following this, the CAFFE technique [41] enhances MobileNetV2 for quicker classification and detection. CAFFE is a deep learning framework that trains and uses deep neural networks fast. Its speed and adaptability make it perfect for many computer vision applications. The foundation of standard MobileNetV2 is the integration of ImageNet [43] and Keras [44] applications into a pre-trained architecture. In-depth descriptions of the layers are provided below:

i. Input (224 × 224 × 3): The images are 224 × 224 pixels in size and have red, green and blue (RGB) colour channels.
ii. Freezing MobileNetV2 Layers: The Standard MobileNetV2 framework depends mainly on feature extraction. MobileNetV2 is a DCNN for image categorization. Importantly, the layer weights are "frozen" to maintain consistency. iii. Transfer learning: It uses a dataset to adapt a model trained for one task to another. This

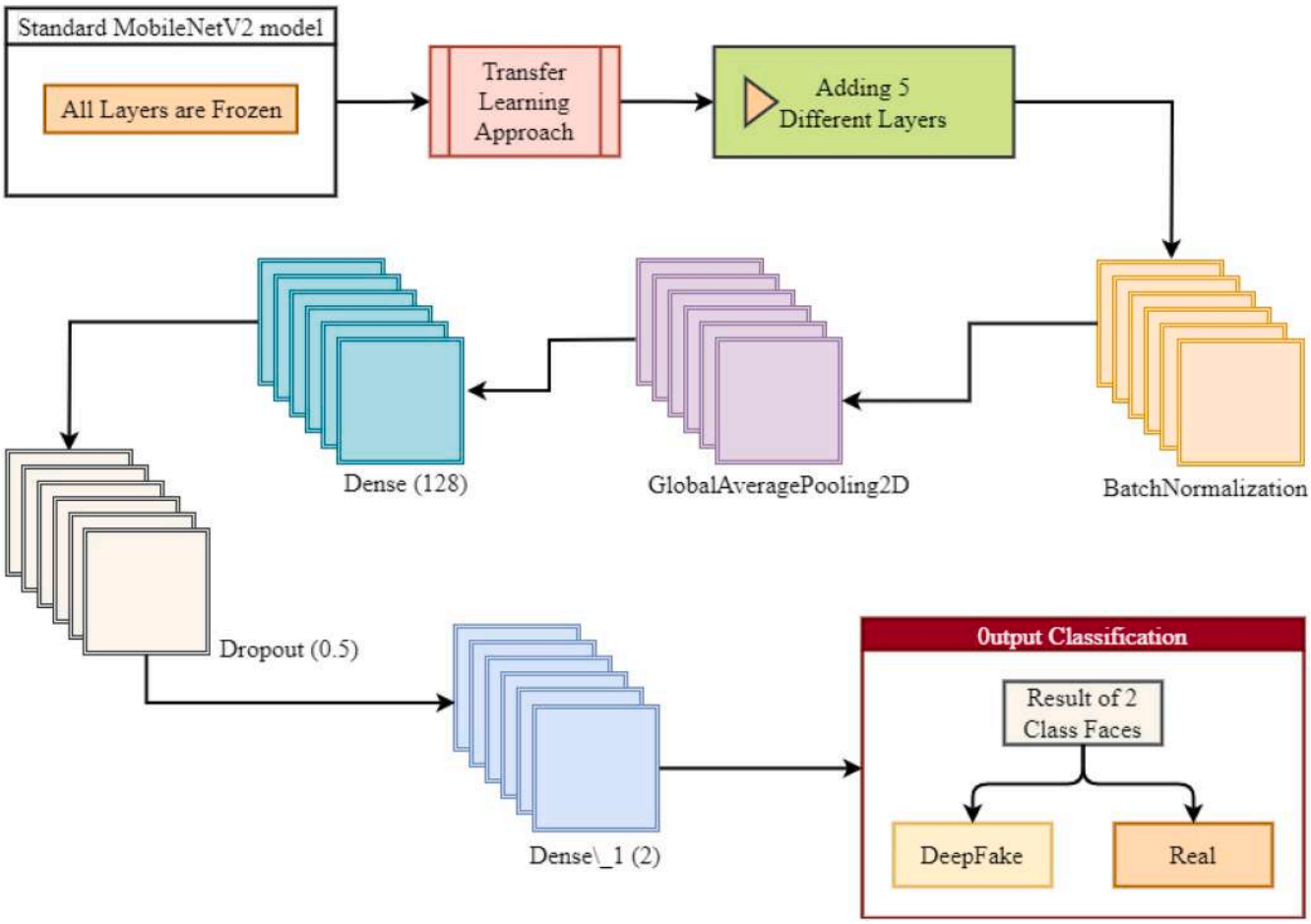


Fig. 2. Customized MobileNetV2 framework.

study uses transfer learning to take advantage of all relevant data without training.

iii. BatchNormalization: This layer normalizes the earlier layer output, improving training progress and stability.
iv. GlobalAveragePooling2D: The feature map's spatial dimensions are blurred by MobileNetV2. GlobalAveragePooling builds a dimensioned feature vector by averaging feature map data. This is crucial because it ensures that input image features are modified evenly at all sizes.
v. Dense (128): This layer's 128 neurons collect complex patterns and features using a specified feature vector.
vi. Dropout (0.5): The dropout rate shows that randomly, 50% of neurons in this layer remain unused after each training cycle to reduce overfitting.
vii. Dense 1 (2): The model's output can categorize all classes, matching the number of neurons in the final dense layer. The output is created by this layer after analyzing results for each of the 2 categories in this research study.

Transfer learning refers to the technique of applying a model trained on one task to a related task that has a smaller dataset. The proposed hybrid CMNV2 model was developed using this approach. Baseline CNN, models are built from the ground up and trained on large volumes of labeled data for specific tasks. In contrast, transfer learning leverages pre-trained machine learning and deep learning models, transferring the knowledge they've acquired from one domain to another. This becomes particularly useful when labeled data is scarce. The CMNV2 model was built by combining the modified MobileNetV2 by incorporating five additional custom layers aimed at improving feature extraction and classification performance specific to deepfake detection. The rationale was to enhance the network's capacity to capture subtle facial manipulations present in deepfakes, which are often missed by standard lightweight models like standard MobileNetV2. In this work, MobileNetV2 and CAFFE, which have been previously trained on large datasets, are utilized for pre-training. These models are capable of extracting general features and patterns, which are then applied to classify real and fake face data, as they have already been optimized for classification and prediction tasks. Explanation on the role and impact of the additional five layers incorporated into the CMNV2 model architecture:

i. Batch Normalization: This layer standardizes the outputs of the preceding layer, improving training efficiency and model stability. It ensures the inputs to each layer maintain a mean close to 0 and variance near 1, enhancing robustness. By computing statistics across mini-batches, it normalizes activations and includes learnable scaling and shifting to increase flexibility.
ii. GlobalAveragePooling2D: After BatchNormalization, a GlobalAveragePooling layer is applied. At the MobileNetV2 stage, this layer reduces the spatial dimensions of feature maps by averaging values across each map, creating a fixed-length feature vector. This step ensures uniformity in feature extraction regardless of input image size, while also reducing overfitting and computational load.
iii. Dense (128 units): This Dense layer contains 128 neurons, responsible for refining the extracted features. It compresses high-dimensional data into a compact 128-dimensional vector while retaining important information. This aids in learning an

efficient and expressive feature representation for downstream processing.

iv. Dropout (0.5): Following Dense, a Dropout layer with a 0.5 rate is introduced. It randomly deactivates 50% of the neurons during each training cycle, forcing the model to be less dependent on specific nodes. This regularization technique enhances generalization and reduces the risk of overfitting.

v. Dense 1 (2 units): The final Dense layer contains two neurons corresponding to the binary classification task (real vs fake). It takes the 128-dimensional representation and outputs class probabilities for the two target categories. This output layer concludes the classification process.

These five additional layers were purposefully chosen to introduce non-linearity, control overfitting and to reveal hidden patterns in the dataset. Various testing configurations were explored to analyze the performance of the enhanced MNV2 model by adjusting the number and type of layers. The results consistently demonstrated that this layered structure improved the model's adaptability, training speed and accuracy when handling complex and variable-rich data.

Input data and pre-processing

The block diagram in Fig. 3 illustrates the usage of photo images for model training and testing. First, data of real and deepfake face images was collected. This study is based on the data collection strategy described in Section 3. Analyzing each person's expression was important. Folders were used to organize the data for future use. A pre-processing program changes the image size to fit the suggested technique for each file.

Data splitting, CMNV2 model training and testing

The dataset contains 5,000 face images, separated into training and testing groups. Section 3 provides a detailed explanation of the dataset's split, which allocates 20% for testing and 80% for training. After loading the baseline model, transfer learning is used to generate all 5 layers, as explained in Section 4.1. New set of trainable CAFFE layers improved facial feature detection and categorization. The model was modified after including CAFFE, using the "weight" files and 2 "TXT" files to preserve dimensions. With minimal processing time, pre-trained models can add bias weights while retaining features. Next, we trained the proposed model for 59 epochs at 0.001 using 32 training batches.

To ensure optimal training performance of the CMNV2 model, we conducted extensive hyperparameter tuning using a grid search strategy. The tuning process was carried out on a validation subset (20% of the dataset) to avoid overfitting and ensure generalization. For the learning rate, we tested values in the range 0.001,0.0001,0.1,0.00001 and observed that a learning rate of 0.001 consistently yielded faster convergence and higher validation accuracy. For batch size, we experimented with 16, 32 and 64 and found that 32 offered the best trade-off between GPU memory usage and gradient stability. The number of epochs was selected based on early stopping criteria. We monitored the validation loss and stopped training once the loss plateaued or began to increase. The model consistently converged within 59 epochs, beyond which no significant gain was observed. Additionally, we used the Adam optimizer with which performed better than SGD in terms of both convergence speed and final accuracy.

In the proposed approach, each video frame is processed using the hybrid CMNV2 model at a frame rate of 30 Frames Per Second (fps), which is widely supported by many systems, devices and broadcasting

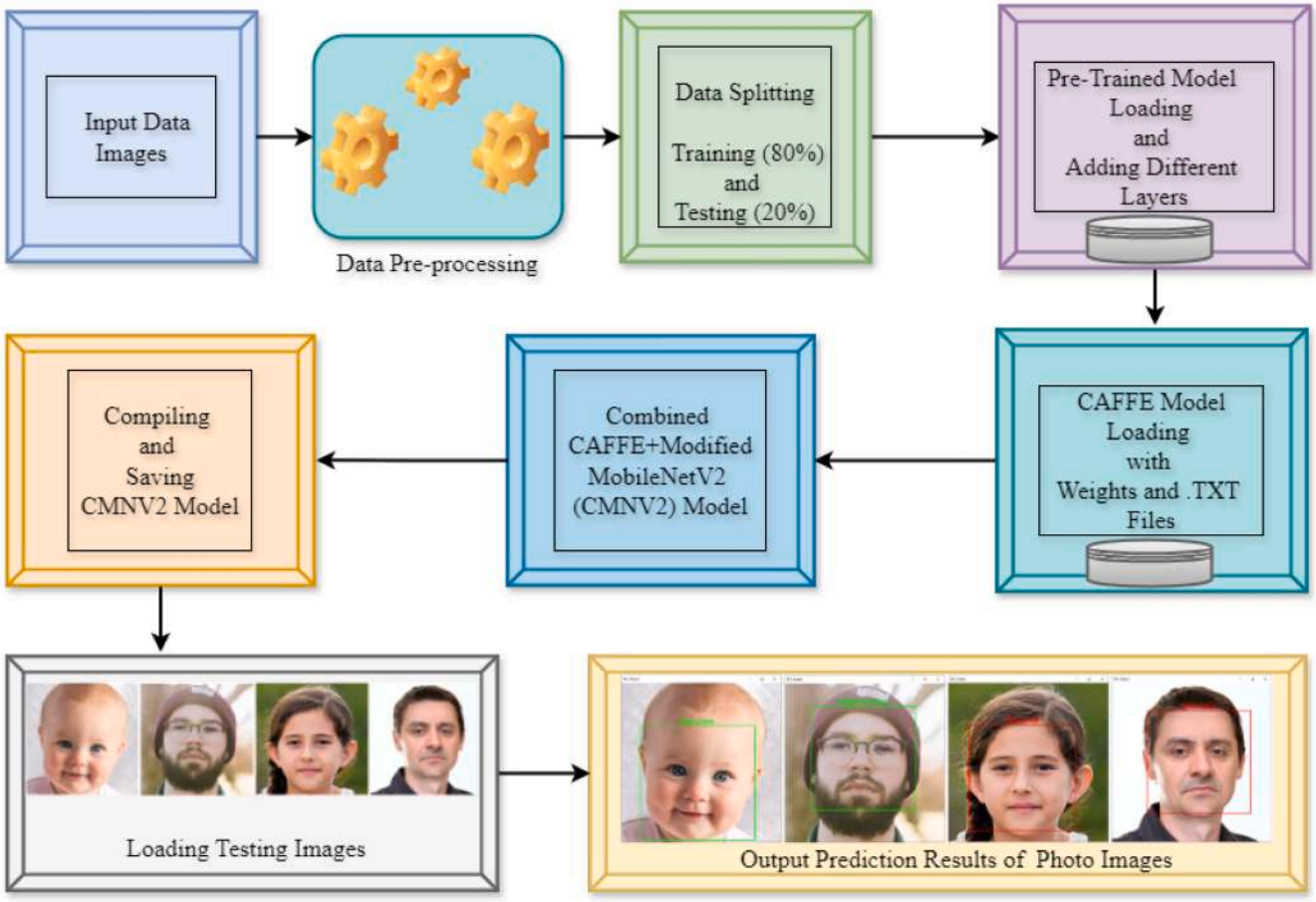


Fig. 3. CMNV2 block diagram.

standards. While higher frame rates like 60 fps require more power for streaming and storage, 30 fps offers a practical balance between smooth performance and visual quality. It ensures consistent video playback, reduces flickering and results in smaller file sizes an advantage for mobile and storage-constrained platforms. This frame rate also helps maintain visual clarity without requiring high-speed connections.

The CMNV2 model was indeed trained and evaluated using static images extracted from the dataset (5,000 real and deepfake facial images), as described in Sections 3 and 4. However, the model was further tested in a real-time video inference scenario, where video streams were processed frame-by-frame. Each frame was treated as an individual image input to the CMNV2 model, enabling the classification of faces as real or deepfake in live or recorded video streams. This use case was demonstrated in Figure 8, where bounding boxes and classification labels were overlaid on video frames. Each video frame is first passed through a face detection step using OpenCV's to extract facial regions, which are then resized to 224×224 and fed into the trained CMNV2 model. The model outputs real or deepfake labels per frame, which are aggregated for visualization only. This confirms the model's operability on video streams.

CMNV2, which combines improved MobileNetV2 with CAFFE, is used for deepfake classification. We distinguish real and deepfake faces using the updated MobileNetV2 framework and the CAFFE framework for FaceNet after training. Deep learning frameworks like CAFFE and OpenCV preprocess and customize images using blob data. The multi-dimensional array format for image storage works well for DNNs. Blobs normalize incoming data before feeding it into a neural network. The computer vision toolkit OpenCV can analyze photos and videos in real-time. The technology analyzes static images and videos using DL. OpenCV's pre-trained face detection and classification models can detect faces in photos and videos. OpenCV finds blobs or zones of interest in photos or frames with faces. Image scaling, normalization and color correction precede neural network processing in OpenCV. The CMNV2 method was suggested for real and deepfake faces. This includes creating a blob function using the image's dimensions to train a DNN for detection and classification. The blob function was then applied to facial features and predicted locations.

We trained the hybrid CMNV2 model with 2 face categories and evaluated its output using photo images. The requirement is accomplished by determining the item's X and Y bounding boxes once its probability of detecting a face in images approaches 0.5. When a prediction is correct, the name class labels are green; otherwise, they are red. This feature keeps bounding boxes inside real-time video and photo boundaries. Labels and borders were created using these resources. DeepFake and real face predictions using weights, blob function, prototxt files, and dimensions of RGB images with labels, as well as boundary boxes, were presented. When prediction is correct, the name class labels are green coloured; otherwise, they are red as shown in Fig. 7 and Fig. 8 for detected real-time video and photo image outputs.

Results and discussion

Proposed hybrid CMNV2 technique uses a deep neural network framework with the OpenCV, TensorFlow and Keras libraries to accurately detect and categorize deepfake and real faces in real-time video and photo images. There are several files in the user's text, including TXT, deploy, prototxt, res10 300×300 ssd iter 140000. caffemodel. The weights of every layer are contained in the CAFFE model, and the layer architecture is detailed in the prototxt and pbtxt files. The experiments in this research were carried out using the Jupyter Notebook, a helpful resource for deep learning model training. Python 3.9 libraries that we utilized in this research, including scikit-learn, pandas and pip, were installed. Pandas, a powerful data processing tool, is used to import and update this dataset and Scikit-learn, an open-source tool for ML and DL models, is used to assess the strategy's effectiveness. For the experiment, a personal computer with 64 GB of random access memory (RAM) and a W-2125 Xeon(R) Intel(R) CPU operating at 4.00 GHz was used.

The accuracy of the CMNV2 model in detecting between real and deepfake faces is 99.10%. In statistical analysis, machine learning and deep learning, the confusion matrix is a useful tool for evaluating the performance of classification models [45]. With 2 potential outcomes, true positives

(T_{+ve}), true negative (T_{-ve}), false positive (F_{+ve}) and false negative (F_{-ve}), this matrix shows the detected and actual categories of a dataset. Number of instances of each class that were accurately or wrongly identified is shown in each column of matrix.

$$\text{Accuracy} = \frac{T_{+ve} + T_{-ve}}{T_{+ve} + F_{+ve} + F_{-ve} + T_{-ve}} = 99.10\% \quad (1)$$

$$\text{ErrorRate} = \frac{F_{+ve} + F_{-ve}}{T_{+ve} + F_{+ve} + F_{-ve} + T_{-ve}} = 0.90 \quad (2)$$

We have extended the experimental evaluation by including five independent runs for each model configuration on all datasets (CelebA, LFW, DFD, and FaceForensics++).

i. For each reported accuracy, precision, recall, and F1-score, we now provide the mean ± standard deviation (SD) values to reflect the consistency and robustness of Hybrid CMNV2 compared to baseline methods.
ii. We have conducted a statistical significance test (paired t-test, p less than 0.05 to compare Hybrid CMNV2 against the strongest baseline (EfficientNetV2 + Xception). Results indicate that the performance improvements of CMNV2 are statistically significant across all datasets.
iii. Additional performance confidence is reported using 95% confidence intervals (CIs), demonstrating that the results are not only superior on average but also stable across multiple runs.

Hybrid CMNV2 achieved an average accuracy of 97.8% ± 0.4% on the FaceForensics++ dataset, compared to 95.9% ± 0.7% by the strongest baseline. The paired t-test confirmed statistical significance ($p = 0.018$).

Performance of 14 different models

Fig. 4 (a) to Fig. 4 (n) show the evaluation of 14 distinct models of confusion matrices evaluated in this research work.

The fewer trainable parameters result in faster training, lower memory usage and reduced risk of overfitting, which improves computational efficiency without decreasing performance.

i. Total parameters: The complete number of parameters (weights and biases) in the model, including both trainable and non-trainable.
ii. Trained parameters: Parameters that are updated during training using back propagation. These are typically found in the custom layers we added.
iii. Non-trained parameters (also called frozen parameters): Parameters that are not updated during training, usually from pre-trained layers retained for feature extraction (e.g., from MobileNetV2 backbone).

Table 1 and Fig. 5 demonstrate that the CMNV2 model outperformed 13 remaining models in terms of accuracy, recall, f1-score, error rate, and precision. The run-time performance measures, including total parameters, trained parameters, and untrained parameters, have been calculated for each model in Table 2.

Compared to the other 13 standard models, the CMNV2 model is more efficient in terms of both architecture and computation. There are 156 non-trainable layers and 5 trainable layers in the CMNV2 model's total of 161 layers. With an accuracy rate of 99.10%, it operates

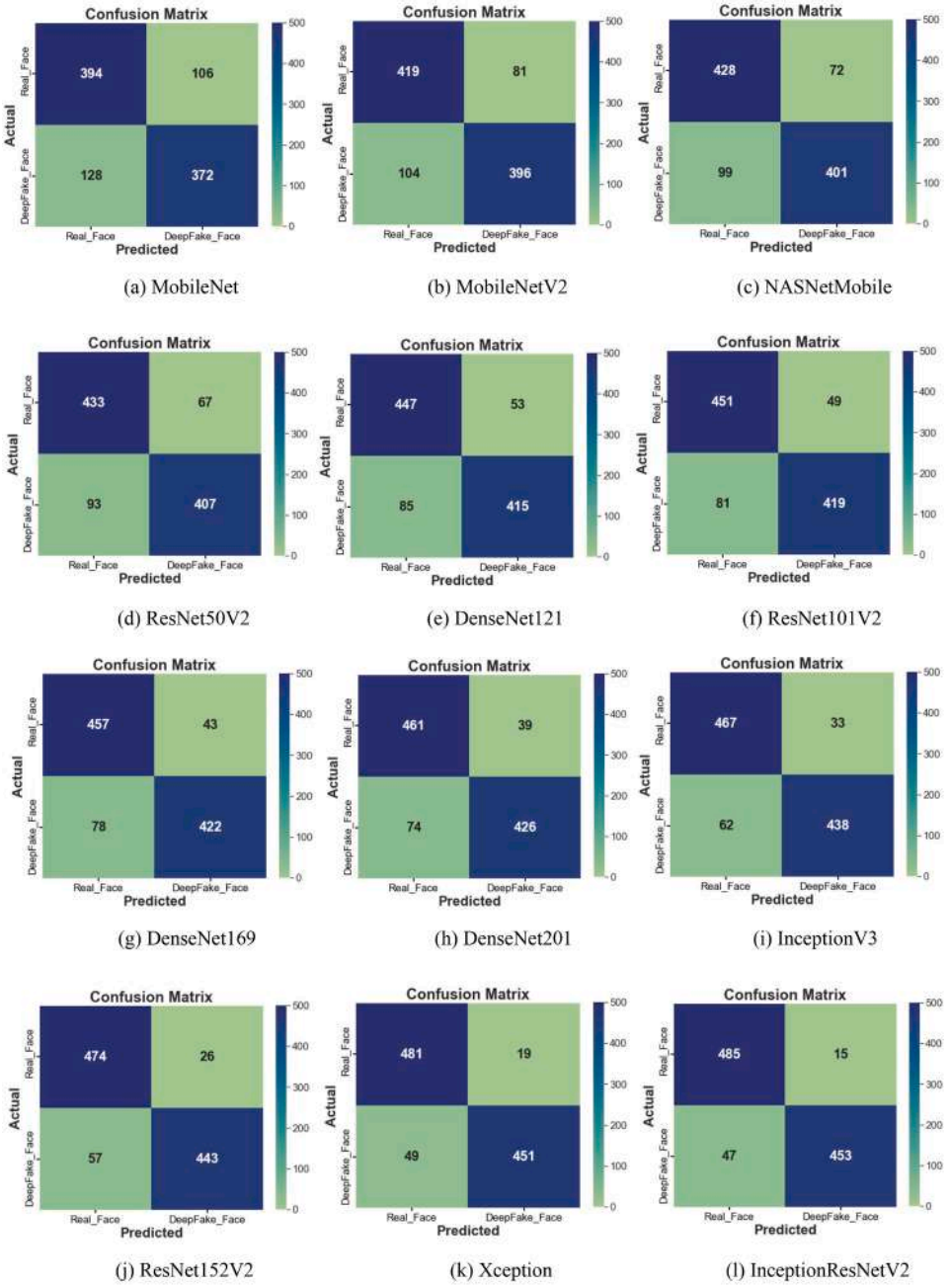


Fig. 4: Evaluation of confusion matrices of 14 models (Continued)

Fig. 4. Evaluation of confusion matrices of 14 models (Continued) Fig. 4: Evaluation of confusion matrices of 14 models.

Table 1
Performance evaluation comparison of 14 models.

Sl.No.	Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Error Rate (%)
1	MobileNet	76.60	75.48	78.80	77.10	23.40
2	MobileNetV2	81.50	80.11	83.80	81.91	18.50
3	NASNetMobile	82.90	81.21	85.60	83.34	17.10
4	ResNet50V2	84.00	82.31	86.60	84.40	16.00
5	DenseNet121	86.20	84.02	89.40	86.62	13.80
6	ResNet101V2	87.00	84.77	90.20	87.40	13.00
7	DenseNet169	87.90	85.42	91.40	88.31	12.10
8	DenseNet201	88.70	86.17	92.20	89.08	11.30
9	InceptionV3	90.50	88.28	93.40	90.76	9.50
10	ResNet152V2	91.70	89.26	94.80	91.95	8.30
11	Xception	93.20	90.75	96.20	93.40	6.80
12	InceptionResNetV2	93.80	91.16	97.00	94.00	6.20
13	EfficientNetV2M	94.80	92.26	97.80	94.95	5.20
14	**Proposed**	**99.10**	**98.23**	**100.00**	**99.10**	**0.90**

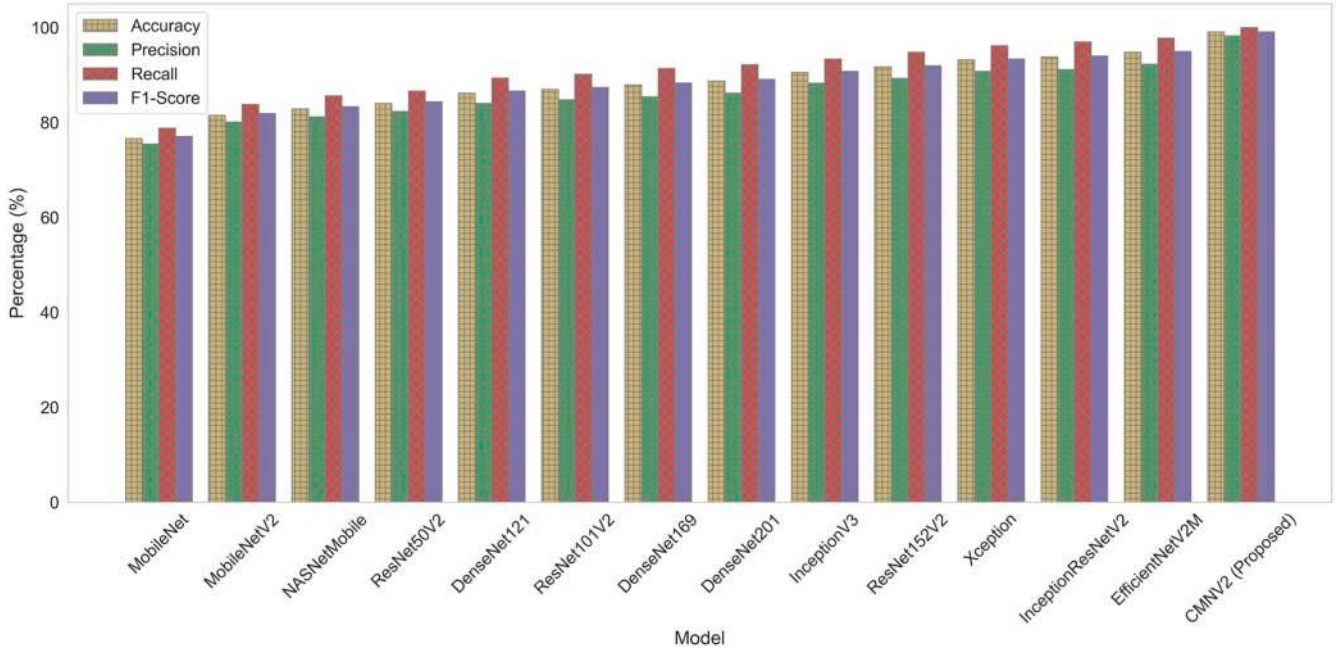


Fig. 5. All 14 models statistical performance analysis.

Table 2
Comparison of 14 different models analysis.

Sl.No.	Model Name	Total Parameters	Trainable Parameters	Non-Trainable Parameters	Run Time (Seconds)
1	MobileNet	32,29,107	32,07,339	21,768	20,509
2	MobileNetV2	22,40,715	22,06,723	33,992	17,606
3	NASNetMobile	53,26,959	52,90,341	36,618	31,084
4	ResNet50V2	6,03,80,891	6,03,37,267	1,43,624	1,74,855
5	DenseNet121	70,37,747	69,54,219	83,528	37,295
6	ResNet101V2	5,26,46,779	5,25,49,235	1,51,240	1,58,660
7	DenseNet169	1,43,08,123	1,41,49,843	1,58,280	46,346
8	DenseNet201	2,02,43,227	2,00,14,291	2,28,936	75,986
9	InceptionV3	2,18,02,927	2,17,68,715	34,312	86,608
10	ResNet152V2	5,83,78,497	5,82,24,633	1,53,864	1,51,355
11	Xception	2,08,82,213	2,08,27,805	54,408	74,642
12	InceptionResNetV2	5,58,74,179	5,58,13,555	60,424	1,68,916
13	EfficientNetV2M	5,31,50,631	5,28,58,719	2,91,912	1,55,910
14	Proposed	25,86,163	3,28,299	22,57,864	3999

accurately. The results of Table 2 and Fig. 6 show that, in terms of parameters and computational time, the CMNV2 model performed better than 13 other models. The following 13 models are explained as follows, which require more computing time as they need more trainable layers: The MobileNet model has 92 trainable layers, 0 non-trainable layers and 92 total layers with an accuracy of 76.60%. At 81.50% accuracy, the

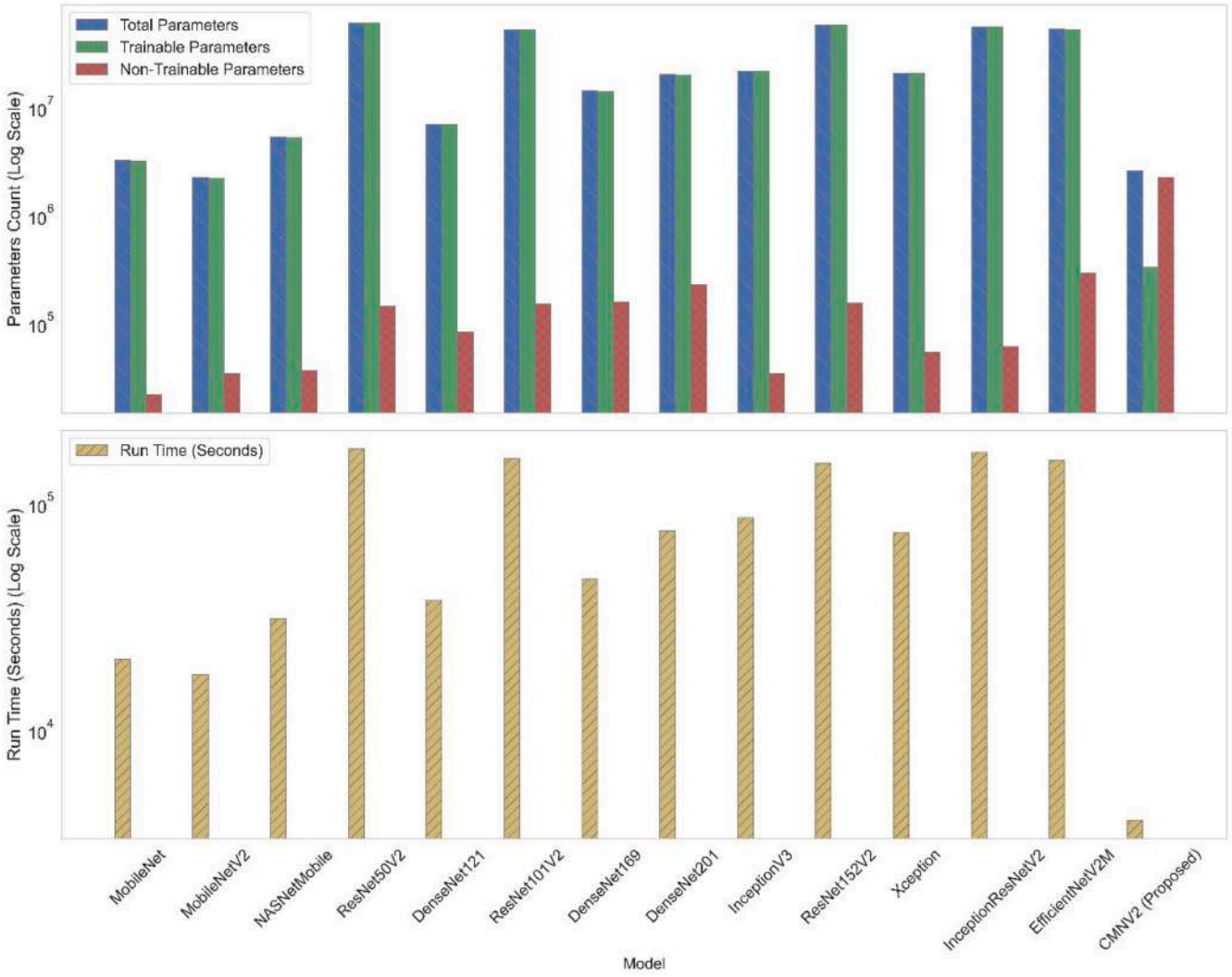


Fig. 6. Log-scaled comparison of model complexity and execution time of 14 models.

MobileNetV2 model has 157 total layers, of which 157 are trainable and 0 are non-trainable. The NASNetMobile model has 772 total layers, 772 trainable layers and 0 non-trainable layers with a performance efficiency of 82.90% accuracy. The ResNet50V2 model contains 192 total layers, 192 trainable layers and 0 non-trainable layers with accuracy performance efficiency is 84.00%. The DenseNet121 model, with 430 trainable layers and 0 non-trainable layers, has an accuracy performance efficiency of 86.20%.

The DenseNet169 model contains 598 total layers, 598 trainable layers, and 0 non-trainable layers, with an accuracy score of 87.90%. The DenseNet201 model contains 710 total layers, of which 710 are trainable and 0 are non-trainable with an accuracy performance efficiency of 88.70%. The InceptionV3 model includes 314 layers in total, 314 trainable layers and 0 non-trainable layers with an accuracy rate of 90.50%. With 567 total layers, 567 trainable layers and 0 non-trainable layers, the ResNet152V2 model has a 91.70% accuracy performance efficiency. With 135 total layers, 135 trainable, and 0 non-trainable, the Xception model contains a 93.20% accuracy performance.

The InceptionResNetV2 model, with 783 total layers, 783 trainable layers and 0 non-trainable layers, achieves an accuracy of 93.80%. The EfficientNetV2M model has 744 trainable layers out of 744 total layers and 0 non-trainable layers with an accuracy performance efficiency of 94.80%.

As the previously discussed challenges demonstrate, the CMNV2 model's different layer design performance and computational efficiency are crucial for achieving the best choice balancing of trainable and non-trainable layers. Out of the 161 total layers, only 5 have been trained; the remaining 156 layers are used, without training and used for feature extraction using a transfer learning technique. With no non-trainable layers and careful layer distribution, the model maintains its robust and stable architecture while learning valuable features. The proposed hybrid CMNV2 model performs much better in terms of architectural characteristics and computing time efficiency when compared to the other 13 conventional models. Table 1 and Table 2 provide a thorough analysis of the run-time performance metrics for each model, including total parameters, trained parameters and Frozen (not updated during training) parameters. Proposed hybrid CMNV2 method has the lowest total trainable parameter values in the category when compared to the other 13 models currently in use, indicating a carefully created architectural design.

The dataset used for training and testing comprised 5,000 images. Real images included 3,000 from CelebA (60% of the real category) and 2,000 from LFW (40% of the real category). Deepfake images consisted of 2,500 each from DFD and FaceForensics++ (50% each of the deepfake category). This proportional distribution ensures a balanced dataset for effective binary classification of real and deepfake faces using the proposed CMNV2 model as shown in Table 3.

Deepfake samples were generated using a combination of manipulation techniques, including FaceSwap, DeepFaceLab, and GAN-based models. The original images used for these manipulations were

Table 3
Proportional breakdown of images from each dataset.

Sl. No.	Category	Dataset Name	Number of Images	Percentage of Category (%)	Percentage of Total Dataset (%)
1	Real Images	CelebA	3,000	60	60
2	Real Images	LFW	2,000	40	40
3	Deepfake Images	DFD	2,500	50	50
4	Deepfake Images	FaceForensics++	2,500	50	50
5	Total		5,000	100	100

sourced from CelebA and LFW datasets. This approach ensured a diverse set of deepfake images varying in facial attributes, backgrounds, lighting conditions, and compression levels, increasing the challenge of binary classification tasks for the proposed CMNV2 model.

Fig. 7 and Fig. 8 illustrate the CMNV2 model's accuracy in predicting 2 face types in real-time video and photos. Faces in Fig. 7 (a) to Fig. 7 (r) are distinguished by the green bounding box that surrounds them with green label with their names when correctly predicted with their names above each box. From Fig. 7 (s) to Fig. 7 (x), faces are incorrectly detected, resulting in red bounding boxes and labels with their names. The green and red box indicates the face's position and if the model detects faces, using a bounding box that indicates the results. Fig. 9 (a) and Fig. 9 (b) display the training and testing performance of the proposed CMNV2 model, prediction accuracy, and loss as 'blue' testing curve and 'green' training curve colours.

A receiver operating characteristic area under the curve (ROC-AUC) shows that the CMNV2 model predicts 2 classes effectively. Fig. 10 shows that the model better distinguishes between the 2 groups, with an outcome larger than 0.5. The convergence of these curves and minimal prediction errors show that the CMNV2 model can distinguish real and deepfake faces. They also help us analyze the model's efficiency and enhance the CMNV2 model's ability to detect faces. Different rising curves for each of the 2 classes reflect model categorization. This curve measures a model's sensitivity and ability to recognize true positives. Another consideration is specificity. It evaluates the ability to identify the model's true negatives. Their consistency and uniformity show that these models are dependable and stable since they predicted every category correctly. The CMNV2 model is adaptive because of its excellent ROC-AUC curve and performance in difficult, binary-class detection of face tasks.

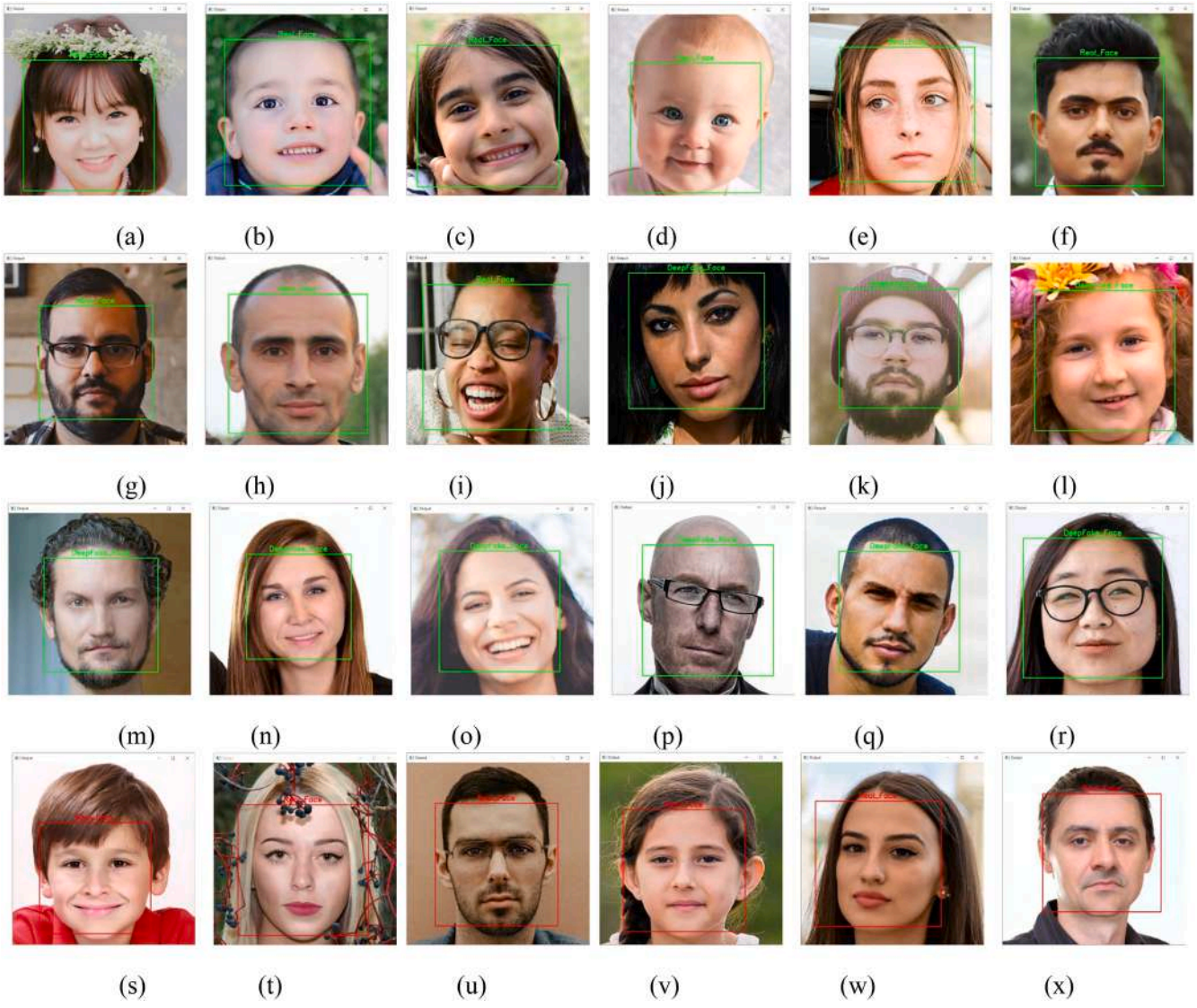


Fig. 7. Output prediction results on photo images.

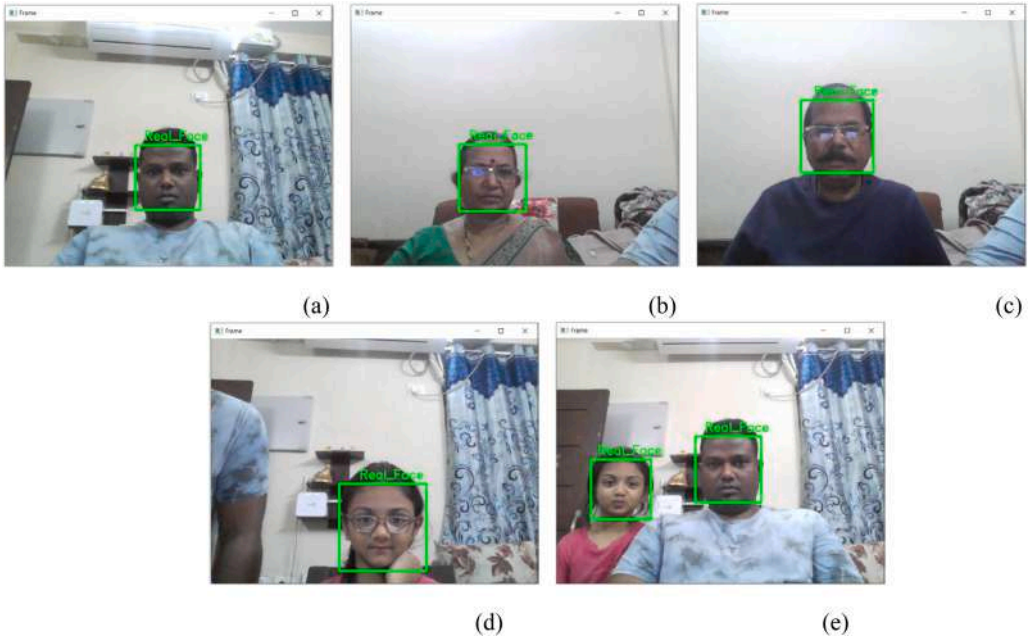


Fig. 8. Output prediction results on real-time video images.

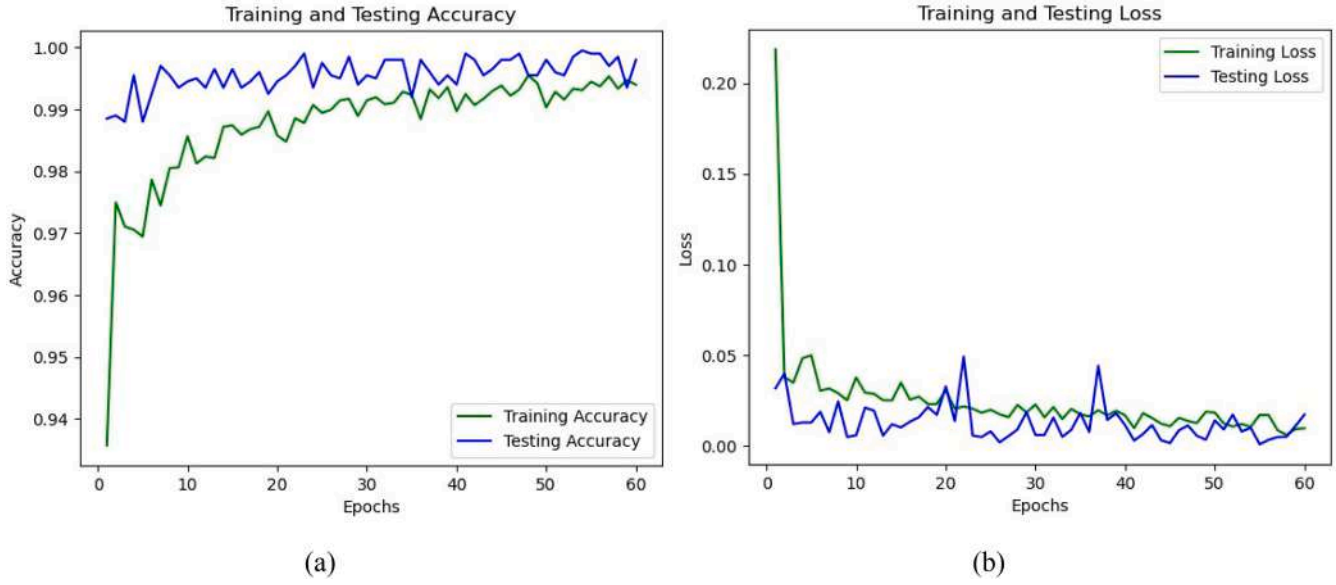


Fig. 9. Accuracy and loss of CMNV2 model.

Conclusion

This article introduced a new hybrid CMNV2 model and the results of real and deepfake facial images. In this work, 5,000 photo images were used. Starting with MobileNetV2, a pre-trained CNN architecture, we fine-tuned the DNN by adding 5 layers using transfer learning. Developed a CAFFE-based deep learning model that increased performance and learning speed. Further, with the combination of these 2 models, a hybrid CMNV2 model was developed and trained on this dataset with 13 other deep-learning methods: MobileNet, MobileNetV2, NASNetMobile, ResNet50V2, DenseNet121, ResNet101V2, DenseNet169, DenseNet201, InceptionV3, ResNet152V2, Xception, InceptionResNetV2, and EfficientNetV2M. The accuracy of 14 evaluated models varied from 76.60% to 99.10%. CMNV2 is better than 13 standard methods with high accuracy (99.10%), low error rate (0.90%), better precision, recall, and f1-score, with less computational time to classify and detect real and deepfake faces. CMNV2 model can be used in various applications, like intelligent attendance systems, healthcare and education.

Limitations: The model performed well on still images but struggled with real-time video, where accuracy drops due to sudden motion, lighting changes, and geographic variations. Rapid participant movement reduces the prediction accuracy, posing a key challenge for video-based applications.

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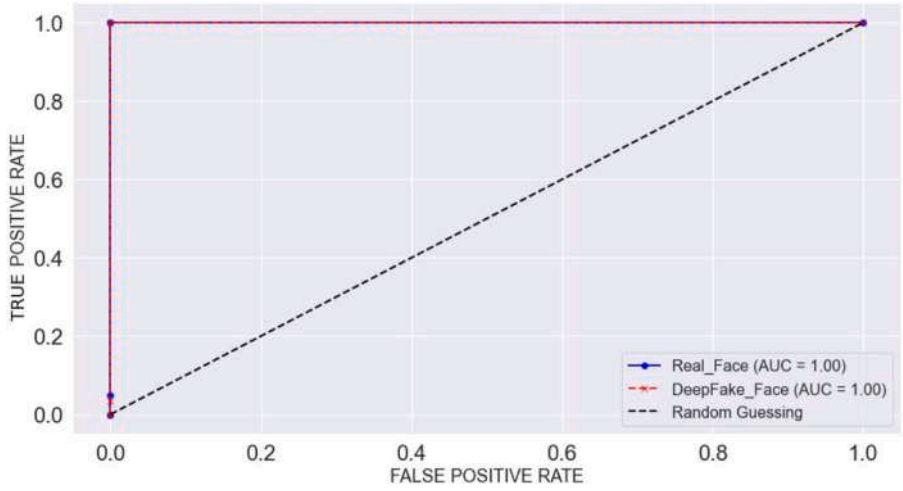


Fig. 10. ROC-AUC curve.

Data Availability

The datasets used in this study are publicly available. Real and fake face datasets were obtained from CelebA, LFW, DeepFakeDetection (DFD), and FaceForensics++ repositories. The original dataset "Real and Fake Face Detection" created by Ciplab@Yonsei University is available on Kaggle (https://www.kaggle.com/datasets/ciplab/real-and-fake-face-detection). Data will be made available upon reasonable request from the corresponding author.

CRediT authorship contribution statement

B. Anil Kumar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Neeraj Kumar Misra:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Nirupma Pathak:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Seyed-Sajad Ahmadpour:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Murugaperumal Krishnamoorthy:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Dhirendra Kumar Shukla:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Mukesh Patidar:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Musawer Hakimi:** Conceptualization, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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